

Present Value Methodology

2025-1

Present Value Concept

- Wealth in Fisher Model:

$$W = Y_0 + Y_1/(1+r)$$

The consumer/producer's wealth is their current endowment plus the future endowment discounted back to the present by the rate of interest (rate at which present and future consumption can be exchanged).

- Why do this?
 - Purpose of comparison—apples to apples (temporal) comparison with multiple agents or apples to apples comparison of investment/consumption opportunities
- Uniform method for valuing present and future streams of consumption in order for appropriate decision making by consumer/producer
- Useful concept for valuing multiple period investments and pricing financial instruments

Calculating Present Value

Present value calculations are the reverse of compound growth calculations:

Suppose V_0 = a value today (time 0)
 r = fixed interest rate (annual)
 T = amount of time (years) to future period

The value in T years we calculate as:

$$V_T = V_0 (1+r)^T \quad \text{(Future Value)}$$

Example

- A \$30,000 Certificate of Deposit with 5% annual interest in 10 years will be worth:
- $V^T = V_0 (1 + r)^T = 30,000 * (1 + 0.05)^{10} =$
 $= \$48,866.84$
- Note: Computation is easy to do in Excel
 $= 30,000 * (1 + 0.05) ^{10}$

Present Value

In reverse:

$$V_0 = V_T / (1+r)^T \quad (\text{Present Value})$$

*The present value amount is the future value discounted
(divided) by the compounded rate of interest*

Example: A \$48,866.84 Certificate of Deposit received
10 years from now is worth today:

$$V_0 = \$48,866.84 / (1+0.05)^{10} = \textbf{\$30,000}$$

Test Review

- Be able to calculate present and future values
- For any three of four variables: (V_0 , r , T , V_T) you should be able to determine the value of the fourth variable.
- How do changes to r and T impact V_0 and V_T ?

r	?
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Example

Q: How many years, T , will it take for an initial investment of V_0 to double if the annual interest rate is r ?

A: Solve $V_0 (1 + r)^T = 2V_0$

$$\Rightarrow (1 + r)^T = 2$$

$$\Rightarrow T \ln(1 + r) = \ln(2)$$

$$\Rightarrow T = \ln(2)/\ln(1+r)$$

$$= 0.69/\ln(1 + r) \approx 0.70/r \text{ for } r \text{ not too big}$$

Present Value of Future Cash Flows

A cash flow is a sequence of dated cash amounts received (+) or paid (-): $C_0, C_1, \dots, C_T$

Cash amounts received are positive; whereas, cash amounts paid are negative

The present value of a cash flow is the sum of the present values for each element of the cash flow

Discount factors: Intertemporal Price of \$1 with constant interest rate of r

$1/(1+r)$ = price of \$1 to be received 1 year
from today $C_i / (1+r)^i$

$1/(1+r)^2$ = price of \$1 to be received 2 years
from today

$1/(1+r)^T$ = price of \$1 to be received T years
from today

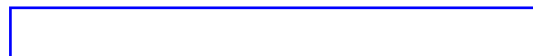
Present Value of a cash flow

- $\{C_0, C_1, C_2, \dots, C_T\}$ represents a sequence of cash flows where payment
- C_i is received at time i . Let r = the interest or discount rate.

Q: What is the present value of this cash flow?

A: The present value of the sequence of cash flows is the sum of the present values:

$$PV = C_0 + C_1/(1+r) + C_2/(1+r)^2 + \dots + C_T/(1+r)^T$$



Summation

$$PV = \sum_{t=0}^T \frac{C_t}{(1+r)^t}$$
$$= C_0 + \sum_{t=1}^T \frac{C_t}{(1+r)^t}$$



Example

You receive the following cash payments:

time 0: -\$10,000 (Your initial investment)

time 1: \$4,000

time 2: \$4,000

time 3: \$4,000

The discount rate = 0.08 (or 8%)

$$\begin{aligned} PV &= -\$10,000 + \$4,000/(1+0.08) \\ &\quad + \$4,000/(1+0.08)^2 + \$4,000/(1+0.08)^3 \\ &= -\$10,000 + \$3,703.70 + \$3,429.36 + \$3,175.33 \\ &= \$308.39 \end{aligned}$$

PV Calculations in Excel

Excel function **NPV**: Net Present Value

`NPV(rate, value1, value2, ..., value29)`

Rate = per period fixed interest rate

value1 = cash flow in period 1

value 2 = cash flow in period 2

...

value 29 = cash flow in 29th period

Note: NPV function does not take account of
initial period cash flow!

Present Value Calculations Short-cuts

PERPETUITY:

A perpetuity pays an amount C *starting next period* and pays this same constant amount C in each period forever:

$C = \text{copon}()$

$$C_1 = C, C_2 = C, C_3 = C, C_4 = C, \dots$$

HMM :: (->) callable bond

$$\begin{aligned} \text{PV(Perpetuity)} &= \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_t}{(1+r)^t} + \dots \\ &= \sum_{t=1}^{\infty} \frac{C_t}{(1+r)^t} = \sum_{t=1}^{\infty} \frac{C}{(1+r)^t} = C \sum_{t=1}^{\infty} \frac{1}{(1+r)^t} \end{aligned}$$

PV of Perpetuity

Based on the infinite sum property, we can write PV as:

$$PV = \text{Initial Term} / [1 - \text{Common Ratio}]$$

$$= C / (1 + r) / [1 - (1 / (1 + r))]$$

$$= C / r$$

$$\text{Initial Term} = C / (1 + r)$$

$$\text{Common Ratio} = 1 / (1 + r)$$

$$\begin{aligned} \text{PV(Perpetuity)} &= C/(1 + r) + C/(1 + r)^2 + C/(1 + r)^3 + \dots \\ &\quad + C/(1 + r)^t + \dots \end{aligned}$$

Let $a = C/(1 + r) = \text{initial term}$



$x = 1/(1 + r) = \text{common ratio}$

Rewriting:

$$\text{PV} = a (1 + x + x^2 + x^3 + \dots) \quad (1.)$$

Post multiplying by x:

$$\text{PV}x = a(x + x^2 + x^3 + \dots) \quad (2.)$$

Subtracting (2.) from (1.):

$$\text{PV}(1 - x) = a \quad \rightarrow \text{PV} = a/(1 - x)$$

$$\text{PV}(1 - 1/(1 + r)) = C/(1 + r)$$

Multiplying through by $(1 + r)$:

$$\text{PV} = C/r$$

Example

The *preferred stock* of a secure company will pay the owner of the stock \$100/year forever, starting next year.

Q: If the interest rate is 5%, what is the share worth?

A: The share should be worth the value to you as an investor today of the future stream of cash flows.

This share of preferred stock is an example of a perpetuity, such that

$$PV(\text{preferred stock}) = \$100/0.05 = \$2,000$$

Example ...

Q: What if the interest rate is 10%?

$$PV(\text{preferred stock}) = \$100/0.10 = \$1,000$$

Notice: That when the interest rate doubled,
the present value of the preferred stock
decreased by $\frac{1}{2}$.

? -> 1/2

Example ...

The preferred stock of a secure company will pay the owner of the stock \$100/year forever, starting this year.

(100)

Q: If the interest rate is 5%, what is the share worth?

A: The share should be worth the value to you as an investor today of the future stream of cash flows (perpetuity component) plus the \$100 received this year.

$$PV(\text{preferred stock}) = \$100 + \$100/0.05 = \$100 + \$2,000 = \$2,100$$

Growing Perpetuity

Suppose the cash flow starts at amount C at time 1, but grows at a rate of g thereafter, continuing forever:

(g) $g=0.1$ 10%

$$C_1 = C, C_2 = C(1+g), C_3 = C(1+g)^2, C_4 = C(1+g)^3, \dots$$

$$\begin{aligned} \text{PV(Perpetuity)} &= \frac{C}{(1+r)} + \frac{C(1+g)}{(1+r)^2} + \frac{C(1+g)^2}{(1+r)^3} + \dots + \frac{C(1+g)^{t-1}}{(1+r)^t} + \dots \\ &= C \sum_{t=1}^{\infty} \frac{(1+g)^{t-1}}{(1+r)^t} \end{aligned}$$

Growing Perpetuity

Based on the infinite sum property, we can write this as:

$$PV = \text{Initial Term} / [1 - \text{Common Ratio}]$$

$$= C/(1 + r)/[1 - ((1 + g)/(1 + r))]$$

$$= C/(r - g)$$

Note: This formula requires $r > g$.

Exercise

- Your next year's cash flow or parental stipend will be \$10,000. Your parents have generously agreed to increase the yearly amount to account for increases in cost of living as indexed by the rate of inflation.
- Your parents have established a trust vehicle such that after their death you will continue to receive this cash flow, so effectively this will continue forever.
- Assume the rate of inflation is 3%.
- Assume the market interest rate is 8%.
- Q: What is the value to you today of this parental support?